Change of Numeraire

Lecture 11 Notes — Units, Measures, Forward Pricing, and FX

Goals and logical structure

Risk-neutral valuation already uses a unit: the money-market account. We change that unit deliberately:

$$\text{positive traded numeraire} \longrightarrow \text{matching measure} \longrightarrow \text{martingale price ratio} \longrightarrow \text{simpler pricing.}$$

The price does not change; only its representation does.

1 Numeraires, measures, and ratios

1.1 The general pricing formula

With money-market account B_t, a payoff X_T is worth

$$V_t = B_t \mathbb{E}^{\mathbb{Q}}\left[\frac{X_T}{B_T}\middle|\mathcal{F}_t\right].$$

Definition 1.1 (Numeraire)**.** A numeraire N_t is a strictly positive traded asset used as the unit of account. The N-denominated price of an asset S is S_t/N_t.

Proposition 1.2 (Change-of-numeraire density)**.** Starting under $\mathbb{Q}$, the measure associated with N is

$$\left.\frac{\mathrm{d}\mathbb{Q}^N}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_t} = \frac{N_t/B_t}{N_0/B_0}.$$

Under $\mathbb{Q}^N$, every traded price divided by N_t is a martingale.

One reusable formula

$$\boxed{V_t = N_t \mathbb{E}^{\mathbb{Q}^N}\left[\frac{X_T}{N_T}\middle|\mathcal{F}_t\right].}$$

Divide the payoff by the numeraire, take expectation under its associated measure, then restore the current unit.

Unit and measure must move together

Dividing by a new numeraire while retaining the old measure generally leaves a drift. The measure change reweights scenarios but does not create or remove possible market scenarios.

Exercise — a bond numeraire

If $N_t = P(t,T)$, what quantity is a martingale and what is the measure called?

Short solution

For every traded asset S, $S_t/P(t,T)$ is a martingale under the T-forward measure $\mathbb{Q}^T$.

1.2 Ratios need Itô corrections

If

$$\frac{\mathrm{d}S_t}{S_t} = \mu_S\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t, \qquad \frac{\mathrm{d}N_t}{N_t} = \mu_N\,\mathrm{d}t + \sigma_N\,\mathrm{d}W_t,$$

then

$$\frac{\mathrm{d}(S_t/N_t)}{S_t/N_t} = \big(\mu_S - \mu_N + \sigma_N^2 - \sigma_S\sigma_N\big)\,\mathrm{d}t + (\sigma_S - \sigma_N)\,\mathrm{d}W_t.$$

The drift is not just $\mu_S - \mu_N$: quadratic variation matters.

2 Forward measures match payment dates

2.1 A bond absorbs discounting

Choose the bond $P(t,T)$. Since $P(T,T) = 1$,

$$\boxed{V_t = P(t,T)\mathbb{E}^{\mathbb{Q}^T}[X_T \mid \mathcal{F}_t].}$$

For a non-dividend-paying asset,

$$F(t,T) = \frac{S_t}{P(t,T)}$$

is a $\mathbb{Q}^T$-martingale.

Example — a zero-value forward

A forward delivering one share for K has value

$$V_t = S_t - KP(t,T) = P(t,T)(F(t,T) - K).$$

If $S_0 = 100$ and $P(0,T) = .95$, the fair delivery price is $F(0,T) = 105.26$.

2.2 A factor-cancelling choice

If $X_T = S_T Y_T$, the stock numeraire yields

$$V_t = S_t\mathbb{E}^{\mathbb{Q}^S}[Y_T \mid \mathcal{F}_t].$$

The candidate must be strictly positive; an arbitrary payoff is not a numeraire.

Exercise — first candidate

A claim pays at T and depends on future bond prices. Which numeraire is the natural first candidate?

Short solution

$P(t,T)$: it matches the payment date and removes discounting from the expectation under $\mathbb{Q}^T$.

3 FX: two currencies, two units

Let X_t be domestic currency per foreign currency unit and

$$B_d(t) = e^{r_d t}, \qquad B_f(t) = e^{r_f t}.$$

The domestic value of the foreign bank account is $X_t B_f(t)$. Therefore, under the domestic measure,

$$\boxed{\frac{\mathrm{d}X_t}{X_t} = (r_d - r_f)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}^d}.}$$

Foreign interest plays the role of a continuous dividend yield.

Example — FX forward

$$F_{0,T} = X_0 e^{(r_d - r_f)T}.$$

For $X_0 = 1.10$ USD/EUR, $r_d = 5\%$, $r_f = 3\%$, and $T = 1$, $F_{0,1} = 1.1222$ USD/EUR.

3.1 FX options

A domestic call on one foreign-currency unit pays $(X_T - K)^+$:

$$C_t = X_t e^{-r_f \tau}\Phi(d_1) - K e^{-r_d \tau}\Phi(d_2),$$

$$d_1 = \frac{\log(X_t/K) + (r_d - r_f + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Parity makes the two discount factors visible:

$$\boxed{C_t - P_t = X_t e^{-r_f \tau} - K e^{-r_d \tau}.}$$

Keep currencies explicit

The spot term represents foreign cash and is discounted at r_f. The strike is domestic cash and is discounted at r_d.

4 Synthesis and practice

Checklist

1. State payoff date and currency.
2. Select a strictly positive traded numeraire.
3. Use its associated measure.
4. Price the normalized payoff and restore the current unit.
5. Check against a forward relation or money-market pricing.

Exercise — final check

Why is it false that a change of numeraire changes the economic price?

Short solution

The claim has one no-arbitrage value. The unit, equivalent probability weights, and martingale ratio change; the economic price does not.